

HDOC-15

Question:

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all digits are even.
- Check whether all digits are prime.

Task1: Algorithm:

Step 1: Start.

Step 2: Declare integer variables num, temp, digit, choice, and flag.

Step 3: Input a positive integer num.

Step 4: Display the menu:

- 1 → Check whether all digits are even.
- 2 → Check whether all digits are prime.

Step 5: Input choice.

Step 6: Copy num into temp.

Step 7: Use switch(choice).

Case 1: Check whether all digits are even

1. Set flag = 1.
2. Repeat while temp > 0:
 - Find last digit: digit = temp % 10.
 - If digit % 2 != 0, set flag = 0 and stop the loop.
 - Remove last digit: temp = temp / 10.
3. If flag == 1, display "All digits are even".
4. Otherwise, display "All digits are not even".

Case 2: Check whether all digits are prime

Prime digits are 2, 3, 5, and 7.

1. Set flag = 1.
2. Repeat while temp > 0:
 - Find last digit: digit = temp % 10.
 - If the digit is not 2, 3, 5, or 7, set flag = 0 and stop the loop.
 - Remove last digit: temp = temp / 10.
3. If flag == 1, display "All digits are prime".
4. Otherwise, display "All digits are not prime".

Step 8: For any other choice, display "Invalid choice".

Step 9: Stop.

Task2: Test case table:

Test Case	Number	Choice	Expected Output	Actual Output	Result
1	12345	1	All digits are not even	All digits are not even	Pass
2	24680	1	All digits are even	All digits are even	Pass
3	24681	1	All digits are not even	All digits are not even	Pass
4	2357	2	All digits are prime	All digits are prime	Pass
5	2379	2	All digits are not prime	All digits are not prime	Pass
6	2468	3	Invalid choice	Invalid choice	Pass

Task3: C program:

```
#include <stdio.h>
int main()
{
    int num, temp, digit, choice, flag;
    printf("Enter a positive integer: ");
    scanf("%d", &num);

    printf("\n1. Check whether all digits are even");
    printf("\n2. Check whether all digits are prime");
    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    temp = num;
    switch(choice)
    {
        case 1:
            flag = 1;
            while(temp > 0)
            {
                digit = temp % 10;
                if(digit % 2 != 0)
                {
                    flag = 0;
                    break;
                }
            }
        }
```

```

        temp = temp / 10;
    }
    if(flag == 1)
        printf("All digits are even");
    else
        printf("All digits are not even");
    break;
case 2:
    flag = 1;
    while(temp > 0)
    {
        digit = temp % 10;
        if(digit != 2 && digit != 3 &&
            digit != 5 && digit != 7)
        {
            flag = 0;
            break;
        }
        temp = temp / 10;
    }
    if(flag == 1)
        printf("All digits are prime");
    else
        printf("All digits are not prime");
    break;
default:
    printf("Invalid choice");
}
return 0;
}

```

Task4: Output:

```
(kali㉿kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 12345

1.Check whether all digits are even
2.Check whether all digits are prime
Enter your choice: 1
All digits are not even
```

```
(kali㉿kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 24680

1.Check whether all digits are even
2.Check whether all digits are prime
Enter your choice: 1
All digits are even
```

```
(kali㉿kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 24681

1.Check whether all digits are even
2.Check whether all digits are prime
Enter your choice: 1
All digits are not even
```

```
(kali㉿kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 2357

1.Check whether all digits are even
2.Check whether all digits are prime
Enter your choice: 2
All digits are prime
```

```
(kali㉿kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 2379

1.Check whether all digits are even
2.Check whether all digits are prime
Enter your choice: 2
All digits are not prime
```

```
(kali㉿kali)-[~/hdoc14.c]
$ ./a.out
Enter a positive integer: 2468

1.Check whether all digits are even
2.Check whether all digits are prime
Enter your choice: 3
Invalid choice
```